

h6 FIR

Algorithm

1 $\Omega = \frac{\omega}{T} \quad \text{IIR}$

$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right) \quad \text{BLT}$

2 $N = \frac{\log\left(\frac{10^{\alpha_s/10} - 1}{10^{\alpha_p/10} - 1}\right)}{2 \log\left(\omega_s/\omega_p\right)}$

$N = \frac{\log\left(\frac{1/\alpha_s^2 - 1}{1/\alpha_p^2 - 1}\right)}{2 \log\left(\omega_s/\omega_p\right)}$

3 $\Omega_c = \frac{\omega_c}{T}$

$\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right)$

$\Omega_c = \frac{\Omega_p}{(10^{\alpha_p/10} - 1)^{1/2N}}$

$\Omega_c = \frac{\Omega_p}{(1/\alpha_p^2 - 1)^{1/2N}}$

4 $p_i = \pm \Omega_c e^{j \frac{N+1-i}{2N} \pi}$

5 $H_d(s) = \frac{\Omega_c^N}{(s - p_1)(s - p_1^*)(s - p_2)(s - p_2^*) \dots}$

6 $T = 1/f_s$
IIR: $s = e^{-p \cdot T} z^{-1}$

BLT: $s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$

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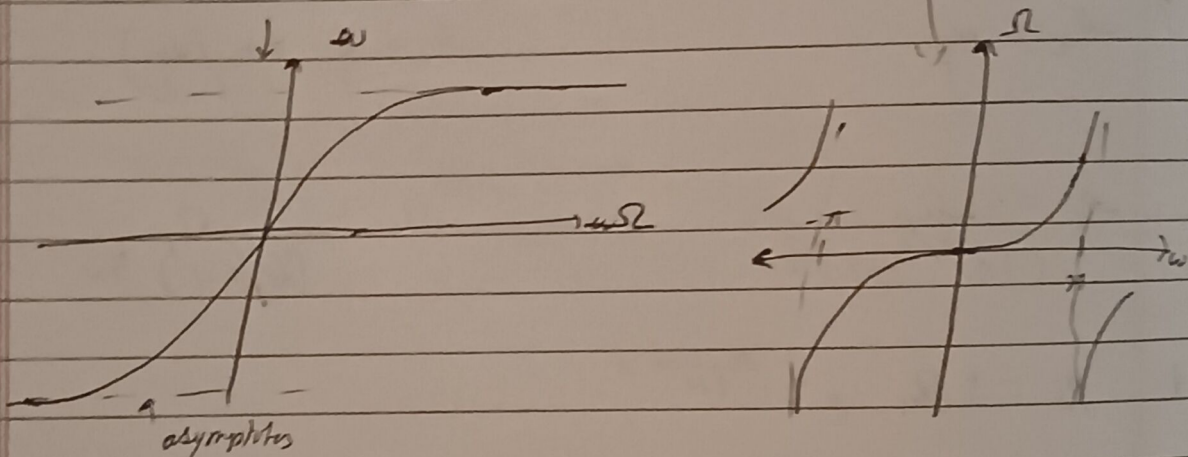
In BLT,

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$

ie. $\tan$ is not a linear fⁿ.

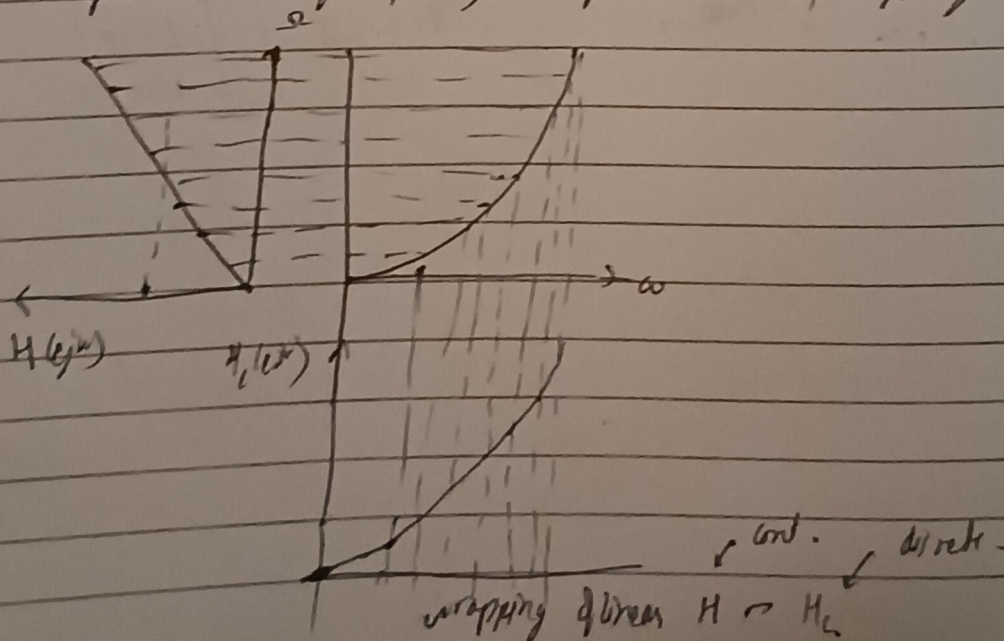
so, when expressing in terms of Ω :

$$\omega = 2 \tan^{-1}\left(\frac{\Omega T}{2}\right)$$



The curve is linear ~~low~~ when $|\Omega| \approx 0$, but at other places, it tends to map highly non linearly.

This phenomenon of frequency compression - Freq Wrapping



PR: 350Hz, -3dB

SR: 1,000Hz, -10dB

$f_s = 5,000$ Hz

LPF: BW, BCT

$$f_{pb} = \frac{350}{5000} = 0.07 \text{ cycles/sample} \quad f_{sb} = \frac{1000}{5000} = 0.2 \text{ cycles/sample}$$

$$\text{so, } f_c = \frac{f_{pb} + f_{sb}}{2} = \frac{0.135 \pi \text{ radians/sample}}{0.135 \text{ cycles}} \quad \text{~~0.135 Hz~~}$$

$$\omega_c = 2\pi f_c = \frac{0.27\pi}{0.135} \text{ radians/sample} \quad \text{~~0.135 \pi radians/sample~~}$$

$$\begin{aligned} Q_c &= \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = 2f_s \tan\left(\frac{\omega_c}{2}\right) \\ &= \text{~~14341.77~~} 4515.17 \text{ rad/sample} \end{aligned}$$

$$N = \frac{\log\left(\frac{10^{4515.17} - 1}{10^{4515.17} - 1}\right)}{2 \log\left(\frac{R_{os}}{R_{op}}\right)}$$

$$\begin{aligned} Q_s &= 2f_s \tan\left(\frac{2\pi f_{sb}}{2}\right) \\ &= 7265.92 \text{ rad/sample} \end{aligned}$$

$$\begin{aligned} Q_p &= 2f_s \tan\left(\frac{2\pi f_{pb}}{2}\right) \\ &= 2235.26 \text{ rad/sample} \end{aligned}$$

$$\begin{aligned} \text{so } N &= \frac{\log\left(\frac{10^{0.1} - 1}{10^{203} - 1}\right)}{2 \log\left(\frac{7265.92}{2235.26}\right)} = 0.545 \\ &\sim 1 \end{aligned}$$

PB: 350 Hz, -9 dB

SB: 1000 Hz, -10 dB

$f_s = 5,000$ Hz

LPF: BW, 0.17

Given

$$f_{PB} = \frac{350}{5000} = 0.07 \text{ cycles / per sample}$$

$$f_{SB} = \frac{1000}{5000} = 0.2 \text{ cycles / per sample}$$

$$\omega_p = 2\pi f_{PB}$$

$$= 0.44 \pi \text{ radians / per sample}$$

$$\omega_s = 2\pi f_{SB}$$

$$= 0.8 \pi \text{ radians / per sample}$$

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right) = 2f_s \tan\left(\frac{\omega_p}{2}\right) = 2.23526 \text{ rad/sample}$$

$$\Omega_s = 2f_s \tan\left(\frac{\omega_s}{2}\right) = 2240.57 \text{ rad/sample}$$

Now,

$$N = \frac{\log\left(\frac{10^{0.5/10} - 1}{10^{0.1/10} - 1}\right)}{2 \log\left(\frac{\Omega_s}{\Omega_p}\right)} = \frac{\log\left(\frac{10^{0.5} - 1}{10^{0.1} - 1}\right)}{2 \log\left(\frac{2240.57}{2.23526}\right)}$$

$$= 0.545 \div 0.939$$

$$= 1$$

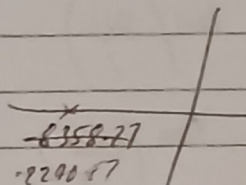
$$\Omega_c = \frac{\Omega_p}{\left(10^{0.5/10} - 1\right)^{1/2N}} = \frac{2.23526}{\left(10^{0.1} - 1\right)^{1/2}} = 8.35827 \text{ rad/sample}$$

$$= 2240.57 \text{ rad/sample}$$

Notes:

$$p_i = \Omega_c e^{j \frac{2\pi f_i n}{2N} \pi}$$

$$\begin{aligned} \text{for } i=0, \quad 2290.57 \\ p_0 &= 8358 e^{j \frac{2\pi \cdot 0}{2N} \pi} \\ &= 8358 e^{j0} \\ &= 8358 \end{aligned}$$



so,

$$H_0(s) = \frac{\Omega_c^N}{s - p_0}$$

location of poles

$$\begin{aligned} &= \frac{10 - p_0}{s - p_0} \\ &= \frac{10 - (-8358.27)}{s - (-8358.27)} \\ &= \frac{10 + 8358.27}{s + 8358.27} \\ &= \frac{2290.57}{s + 2290.57} \end{aligned}$$

so BLT: $s = z \frac{1-z^{-1}}{1+z^{-1}}$

$$\begin{aligned} \therefore H_0(z) &= \frac{2290.57}{10,000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2290.57} \\ &= \frac{2290.57 + 2290.57 z^{-1}}{10,000 - 10,000 z^{-1} + 2290.57 + 2290.57 z^{-1}} \\ &= \frac{2290.57 + 2290.57 z^{-1}}{12290.57 - 7759.43 z^{-1}} \end{aligned}$$

H 82 B a

$$0.8 \leq |H(e^{j\omega})| \leq 1 \quad ; \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.2 \quad ; \quad 0.6\pi \leq \omega$$

$$f_s = 100$$

so,

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$$

$$= 2f_s \tan\left(\frac{\omega_p}{2}\right)$$

$$= 19.99 \text{ rad/sample}$$

$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right)$$

$$= 2f_s \tan\left(\frac{\omega_s}{2}\right)$$

$$= 275.27 \text{ rad/sample}$$

$$N = \frac{\log\left(\frac{\gamma_{dp}^2 - 1}{\gamma_{ds}^2 - 1}\right)}{2 \log\left(\frac{\Omega_s}{\Omega_p}\right)}$$

$$= 1.199$$

$$= 2$$

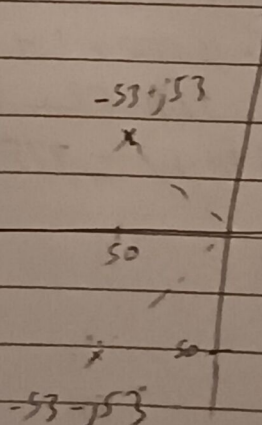
$$\Omega_c = \frac{\Omega_p}{(\gamma_{dp}^2 - 1)^{1/(2N)}} = 75 \text{ Hz rad/sample}$$

the poles

$$p_i = -\Omega_c e^{j \frac{N+2i+1}{2N} \pi}$$

$$\begin{aligned} p_0 &= 75 e^{j \frac{2+1}{4} \pi} \\ &= 75 e^{j \frac{3}{4} \pi} \\ &= -53 + j53 \end{aligned}$$

$$\begin{aligned} p_1 &= 75 e^{j \frac{2+2+1}{4} \pi} \\ &= 75 e^{j \frac{5}{4} \pi} \\ &= -53 - j53 \end{aligned}$$



location of poles

$$\begin{aligned}
 H_d(s) &= \frac{\Omega_c^N}{(s - p_0)(s - p_1)} \\
 &= \frac{5625}{(s + 53 - j53)(s + 53 + j53)} \\
 &= \frac{5625}{s^2 + 2 \times 53s + 75^2} \\
 &= \frac{5625}{s^2 + 106s + 5625}
 \end{aligned}$$

$$\begin{aligned}
 * \quad s &= \frac{z}{T} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) \\
 &= 2f_s \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right)
 \end{aligned}$$

$$\begin{aligned}
 H(z) &= \frac{5625}{\left[200 \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) \right]^2 + 106,200 \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) + 5625} \\
 &= \frac{5625 (1 + z^{-1})^2}{40000 (1 - 2z^{-1} + z^{-2}) + 21200 (1 - z^{-2}) + 5625 (1 + 2z^{-1} + z^{-2})} \\
 &= \frac{5625 + 11250 z^{-1} + 5625 z^{-2}}{6825 - 68750 z^{-1} + 24925 z^{-2}}
 \end{aligned}$$

80 Bd

LRF to HPF:

$$z^{-1} \rightarrow \frac{z^{-1} - a}{1 - a z^{-1}}$$

$$a = \frac{\sin\left(\frac{\omega_c - \omega_c'}{2}\right)}{\sin\left(\frac{\omega_c + \omega_c'}{2}\right)}$$

ω_c = old filter band edge frequency

ω_c' = new filter band edge frequency

if LRF: $\frac{12240.57}{s+1}$

$$\frac{12240.57 + 12240.57 z^{-1}}{12240.57 - 12240.57 z^{-1}} \rightarrow \text{filter with } \omega_c = 0.49\pi$$

and say, HPF conversion to $\omega_c' = 0.5$:

$$a = \frac{\sin\left(\frac{0.49 - 0.5}{2}\right)}{\sin\left(\frac{0.49 + 0.5}{2}\right)} = -0.066$$

$$\text{so, new filter: } \frac{12240.57 + 12240.57 z^{-1} + 0.066}{12240.57 - 12240.57 z^{-1} - 0.066 z^{-1}}$$

78 B4

PB : 5dB, 200Hz

SB : 12dB, 500Hz

 f_s : 5000Hz

$$\omega_p = \Omega_p = \frac{\omega_p}{T} = \omega_p f_s$$

$$= 2\pi \times 200 \times 5000$$

$$= 2000000\pi$$

$$\Omega_s = \omega_s f_s$$

$$= 2\pi \times 500 \times 5000$$

$$= 5000000\pi$$

$$N = \frac{\log \left(\frac{10^{a_s/10} - 1}{10^{a_p/10} - 1} \right)}{2 \log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$= \log \left(\frac{10^{12} - 1}{10^5 - 1} \right) = 2.99$$

$$= 2.05$$

$$= 2$$

$$\Omega_c = \frac{\Omega_p}{(10^{a_p/10} - 1)^{1/2N}} = 1,649,309\pi \text{ rad}$$

$$p_1 = \Omega_c e^{j \frac{N+1}{2N} \pi}$$

$$p_0 = \Omega_c (-0.707 + j0.707)$$

$$p_1 = \Omega_c (-0.707 - j0.707)$$

$$\omega, H_0(s) = \frac{\Omega_c^2}{s^2 + 1.414\Omega_c s + \Omega_c^2}$$

Now,

$$\frac{1}{s-pT} \rightarrow \frac{1}{1-e^{-pT}z^{-1}}$$

$$\begin{aligned} \text{So, } H_0(s) &= \frac{\Omega_c^2}{s^2 + 1.414\Omega_c s + \Omega_c^2} \\ &= \frac{\Omega_c^2}{s^2 + 2 \times 0.707\Omega_c s + (0.707\Omega_c)^2 + \Omega_c^2 - (0.707\Omega_c)^2} \\ &= \frac{\Omega_c^2}{(s + 0.707\Omega_c)^2 + 0.5\Omega_c^2} \\ &= \frac{0.707\Omega_c^2 + 0.293\Omega_c^2}{0.5\Omega_c^2 + 0.5\Omega_c^2} \\ &= \frac{(s + 0.707\Omega_c)^2 + (0.707\Omega_c)^2}{0.707\Omega_c \times 1.414\Omega_c} \\ &= \frac{1.414\Omega_c \times 0.707\Omega_c}{(s + 0.707\Omega_c)^2 + (0.707\Omega_c)^2} \end{aligned}$$

in form of:

$$\frac{\Gamma \times \beta}{(s+\alpha)^2 + \beta^2} \rightarrow \frac{\Gamma \times \frac{e^{-\alpha T} (\sin \beta T)}{1 - 2e^{-\alpha T} (\cos \beta T) z^{-1} + e^{-2\alpha T} z^{-2}}}{1 - 2e^{-\alpha T} (\cos \beta T) z^{-1} + e^{-2\alpha T} z^{-2}}$$

$H_0(z) =$

$$\text{So, } = 1.414\Omega_c \times \frac{e^{-0.707\Omega_c T} (\sin \beta T) z^{-1}}{1 - 2e^{-0.707\Omega_c T} (\cos \beta T) z^{-1} + e^{-1.414\Omega_c T} z^{-2}}$$

$$T = 1 = 2 \times 10^{-4}, \quad 0.707\Omega_c T = 366.33$$

$$1.414\Omega_c T = 732.66$$

$$\therefore H(z) = \frac{732.66 \times e^{-366.33} (\sin 366.33) z^{-1}}{1 - 2 \times e^{-366.33} (\cos 366.33) z^{-1} + e^{-732.66} z^{-2}}$$

pre-warping is necessary to get a linear frequency mapping as $\tan^{-1}$ is non linear in BZT method.

Chebyshev form not given, so assuming Chebyshev-1.

lot, forget & did IIR method.

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$$\omega_p: 0.707, 0.2\pi, f_s = 5000$$

$$\omega_s: 0.1, 0.5\pi, \Omega_s = \frac{\omega_s}{T} = \omega_s F = 5000\pi$$

$$\text{ripple parameter, } \epsilon = \left(\frac{1}{\alpha_p^2} - 1 \right)^{0.5} = 1 \quad \omega_p - \omega_s F = 2000\pi$$

order of filter:

$$|H(\Omega)| = -20 \log(\epsilon) - 6(N-1) - 20N \log(\Omega_s)$$

$$\text{at } \Omega = \Omega_s, |H(\Omega_s)| = -20 \log(0.1) = 20 \text{ dB}$$

so,

$$-20 = -20 \log(1) - 6N + 6 - 20N \log(5000\pi)$$

$$-20 = -6N + 6 - 89.922N$$

$$N = \frac{576.922}{-89.922} = 6.41$$

$$= 0.289$$

$$\sim 1$$

Now,

$$\theta_i = \frac{N+2i+1}{2N} \pi$$

$$\theta_0 = \frac{2}{42} \pi$$

$$= \pi$$

$$x_i = r \cos \theta_i$$

$$y_i = R \sin \theta_i$$

$$r = \frac{2\beta}{\beta^2 - 1}, R = \frac{2\beta}{\beta^2 + 1}$$

$$\beta = \left[\frac{\sqrt{1+\epsilon^2} + 1}{\epsilon} \right]^{1/N}$$

$$= 2.914$$

$$r = \frac{2000\pi \times 2.914^2 - 1}{1 \times 2.914}$$

$$= 2000\pi$$

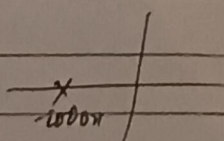
$$R = \frac{2000\pi \times 2.914^2 + 1}{1 \times 2.914}$$

$$= 2828.93\pi$$

$$x_i = 2000\pi \cos(n) = -2000\pi$$

$$y_i = 2828.93\pi \sin(n) = 0$$

So, pole:



TF:

$$H_u(s) = \frac{c}{\prod_{i=0}^{N-1} (s - p_i)}$$

$$c = \begin{cases} b_0 & \text{if } N \text{ is odd} \\ \frac{b_0}{\sqrt{1+\epsilon^2}} & \text{if } N \text{ is even} \end{cases}$$

$$\therefore c = b_0$$

$$H_u(s) = \frac{b_0}{s + 2000\pi}$$

$\therefore b_0 = 2000\pi$, last (constant) term

$$H_u(s) = \frac{2000\pi}{s + 2000\pi}$$

in digital:

$$H(z) = \frac{2000\pi}{1 - e^{2000\pi T} z^{-1}}$$

$$= \frac{2000\pi}{1 - e^{0.9\pi} z^{-1}}$$